Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/28/2008
PI Name:	Brady, Joseph V. Ph.D.		
Project Title:	Psychosocial Performance Factors in Space Dwelling Groups		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	JVB@jhmi.edu	**Fax:**	FY 410-550-2780
PI Organization Type:	NON-PROFIT	**Phone:**	410-550-2800
Organization Name:	Institutes for Behavior Resources, Inc.		
PI Address 1:	Psychiatry and Behavioral Science		
PI Address 2:	The Johns Hopkins School of Medicine		
PI Web Page:			
City:	Baltimore	**State:**	MD
Zip Code:	21224-6823	**Congressional District:**	3
Comments:	NOTE: PI Dr. Joseph Brady passed away in July 2011.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	04/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 4/30/2008 per K. Major/NSBRI (3/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hienz, Robert (The Johns Hopkins University School of Medicine) Hursh, Steven (Institutes for Behavior Resources, Inc.)		
Grant/Contract No.:	NCC 9-58-NBPF00403		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

The original aims of this project focused upon the development of an experimental test bed for modeling performance effectiveness and psychosocial adaptation in support of exploratory spaceflight missions beyond Earth's atmosphere. The research methodology adopted to accomplish these objectives provided for distributed interactive multi-person simulations in computer-generated environments focused upon psychosocial performance interactions within and between space-dwelling flight crews and Earth-based control center groups. The experimental setting provided the automated means for setting the context for the analysis of space crew performance as well as monitoring electronically the interactive effects of simulated communication modality constraints and other environmentally stressful conditions. Distributed interactive simulation experiments characterized the extent to which variations in the structure and function of communication channels within and between space-dwelling and Earth-based groups affect the ability to accurately discern and respond to the realities and requirements of mission operations. The effects of environmentally stressful time constraints as well as the appetitive and aversive characteristics of programmed incentive conditions have been analyzed in the context of spaceflight simulations.

The results of these distributed interactive simulation studies showed clearly that cooperative and productive interactions were maintained between individually isolated and dispersed crewmembers in a task-driven environment. All experimental flight crews actively engaged in communicating and effective problem-solving over extended time intervals without benefit of one another's physical presence. In addition, investigations of communication modality constraints indicated that there was a high degree of interchangeability between available communication modes as well as use of 'indexical' features of the simulation display as countermeasures to maintain effective crew performance. Further, variations in crew configurations (crewmates reassigned to vehicles other than those normally operated) showed clearly that performance effectiveness levels were significantly reduced during missions in which crew assignment changes resulted in inexperienced, or "Weak" crew configurations, as contrasted with highly-experienced, or "Strong" crew configurations. Crew composition variations were also shown to have effects upon group dynamics that were not necessarily reflected by the crew performance measures. Finally, studies of positive and negative incentive conditions on crew performance effectiveness have shown increased performance effectiveness under both positive and negative incentive conditions, but an adverse change in psychosocial adaptation – i.e., negative incentive conditions characterized by a pronounced increase in negative self-report ratings accompanied by decreased positive self-report ratings, and positive incentive conditions characterized by positive self-report ratings only. Thus incentive conditions associated with simulated spaceflight missions can significantly affect psychosocial adaptation without compromising task performance effectiveness in trained and experienced volunteer crews.

Newly-initiated studies on management system variations consisted of developing methodologies for assessing crew performance effectiveness under rigid schedule-based management of crew activities by Mission Control versus more flexible, contingency-based, autonomous management of crew activities by themselves. Autonomous management consisted of having a crew arranging all aspects of their exploratory activities. Schedule-based management was simulated by Mission Control-defined performance requirements with regard to ordering and time spent in each geographic region and surface vehicle task assignments, and requiring that this schedule be strictly adhered to. Tests of this methodology have shown clear adverse effects of mission control-defined schedule limitations that resulted in both decrements in crew performance effectiveness and increases in negative psychosocial adaptation self-report measures.

Assessment of crew performance and psychosocial adaptation has also been extended to longer-duration, 12-hour mission simulations conducted during different parts of a 24-hour circadian cycle (either 'Early': 9 am – 9 pm, or 'Late': 9 pm – 9 am). 'Late' workloads produced significant decrements in individual performances that included reaction time deficits, attentional lapses, psychosocial communication reductions, failures to properly report rule solutions, and inadvertent discarding of high-valued geologic samples. Significant changes occurred in the patterns of positive, negative, and physical self-report ratings following 'Late' workload Missions such that positive self-reports decreased while negative and physical self-reports markedly increased. Importantly, in spite of these numerous decrements, crew performance efficiency remained at normal levels. Lapses in individual crewmember performances thus appeared to be compensated for in the overall crew performance, indicating the utility of multi-person crews as countermeasures to failures or reductions in individual performances under adverse workload conditions. Nevertheless, the dramatic decrease in psychosocial communication during late missions could be an early warning sign of crew desynchronization that could hinder crew cohesion and adaptability to malfunctions, environmental disruptions, or schedule demands.

Future plans involve migration of the simulation scenario providing for a broader range of Lunar surface regions to enhance the complexity of geologic specimen identification, collection and conceptual analysis task. Variations in core communication functions under this more complex model will continue to focus upon an experimental analysis of the interactions between ground-based control and flight crew performance effectiveness.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The methodological development associated with this NSBRI-funded research has provided a test bed for modeling performance effectiveness and psychosocial adaptation in computer-generated distributed interactive multi-person environments. This experimental setting provides an automated means for setting the task performance context as well as monitoring electronically the interactive effects of communication modality constraints and other environmentally stressful conditions. Such a distributed interactive experimental analysis can characterize the extent to which variations in the structure and function of communication channels within and between individuals and groups affect the ability to accurately discern and respond to the realities and requirements of both operational and interpersonal transactions. The effects of environmentally stressful time constraints, appetitive and aversive characteristics of programmed incentive conditions, workload conditions, and management system conditions can be analyzed experimentally in the context of such distributed interactive environments.

Research conducted within the context of this distributed interactive simulation model can provide the basis for developing effective patterns of communication and problem solving strategies as well as a range of training procedures to enhance problem solving effectiveness. The Earth benefits to be derived from applications of this research will extend to the small operational group selection and training process as well as the management of stressful interactions as well as the maintenance of group cohesion and productivity. Not only can the outcome of these studies be expected to have an important impact on safety and the quality of life in many Earth-based applied settings, but larger societal units will ultimately benefit from the resulting conceptual and methodological advances that effectively promote social and ecological stability while concurrently enhancing an education and training technology that assures effective communication of an expanded generalizeable knowledge base.

Task Progress:	Task progress included newly-initiated studies on the effects of management system variations and workload condition variations undertaken during the current reporting year. The newly-initiated studies on management system variations consisted of developing methodologies for assessing crew performance effectiveness under rigid schedule-based management of crew activities by Mission Control versus more flexible, contingency-based, autonomous management of crew activities by themselves. Autonomous management consisted of having a crew arranging all aspects of their exploratory activities, while schedule-based management was simulated by Mission Control-defined performance requirements with regard to ordering and time spent in each geographic region and surface vehicle task assignments, and requiring that this schedule be strictly adhered to. Tests of this methodology have shown clear adverse effects of mission control-defined schedule limitations that resulted in both decrements in crew performance effectiveness and increases in negative psychosocial adaptation self-report measures. Newly-initiated studies on workload condition variations were designed to assess crew performance and psychosocial adaptation under longer-duration, 12-hour mission simulations conducted during different parts of a 24-hour circadian cycle (either 'Early': 9 am – 9 pm, or 'Late': 9 pm – 9 am). 'Late' workloads produced significant decrements in individual performances that included reaction time deficits, attentional lapses, psychosocial communication reductions, failures to properly report rule solutions, and inadvertent discarding of high-valued geologic samples. Significant changes occurred in the patterns of positive, negative, and physical self-report ratings following 'Late' workload Missions such that positive self-reports decreased while negative and physical self-reports markedly increased. Importantly, in spite of these numerous decrements, crew performance efficiency remained at normal levels. Lapses in individual crewmember performances thus appeared to be compensated for in the overall crew performance, indicating the utility of multi-person crews as countermeasures to failures or reductions in individual performances under adverse workload conditions. Nevertheless, the dramatic decrease in psychosocial communication during late missions could be an early warning sign of crew desynchronization that could hinder crew cohesion and adaptability to malfunctions, environmental disruptions, or schedule demands.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Hienz RD, Brady JV, Hursh SR, Banner MJ, Gasior ED, Spence KR. "Communication constraints, indexical countermeasures and crew configuration effects in simulated space-dwelling groups." Acta Astronautica. 2007 Feb-Apr;60(4-7):362-78. http://dx.doi.org/10.1016/j.actaastro.2006.09.005 , Feb-2007
Articles in Peer-reviewed Journals	Hienz RD, Brady JV, Hursh SR, Gasior ED, Spence KR, Emurian HH. "Effects of incentives on psychosocial performance in simulated space-dwelling groups." Acta Astronautica 2008 Oct-Nov;63(7-10):800-10. http://dx.doi.org/10.1016/j.actaastro.2008.03.003 , Oct-2008
Papers from Meeting Proceedings	Gasior ED, Spence KR, Hienz RD, Hursh SR, Brady JV. "Distributed interactive simulation in space dwelling groups." Huntsville Simulation Conference 2007, Huntsville, AL, October 31-November 1, 2007. Proceedings of the Huntsville Simulation Conference, November 2007. , Nov-2007